COSC325 Project Proposal

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I. Problem and Motivation

Here, we propose our final project for COSC325 - Introduction to Machine Learning. We shall use various machine learning methods - such as linear regression, decision trees, and random forest - to predict housing prices based on a collection of features. Our goal is to understand the benefits and drawbacks of different machine learning models first hand.

We are using the the "House Prices - Advanced Regression Techniques" from Kaggle. This data set is appropriate because it has a large set of features and samples - much higher than any other data set we've used in class or on homework assignments. This way, we will can also practice feature selection and extraction where it might be practical.

II. Dataset and EDA

The data used is an advanced housing dataset sourced from kaggle. In this dataset we use log transformation to normalize the distribution 1, compressing the range of high value outliers and expanding the range of lower values. This step is important for linear regression models.

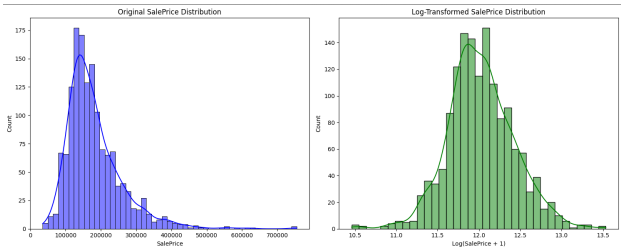

Fig. 1. Normalization of SalePrice

A. Class Balance

The majority of houses are priced between $100k and $250k, 2 with fewer samples below and above this range. This can cause prediction for houses less than $100k and greater than $300k to be less reliable.

B. Missing Values

For missing data, features that indicate absence, valued "NaN" in features: "PoolQC", "Fence", etc... we fill these with "None" to distibguish houses with and without these features. We use context for other features such as "LotFrontage" by entering the median Frontage of the dataset.

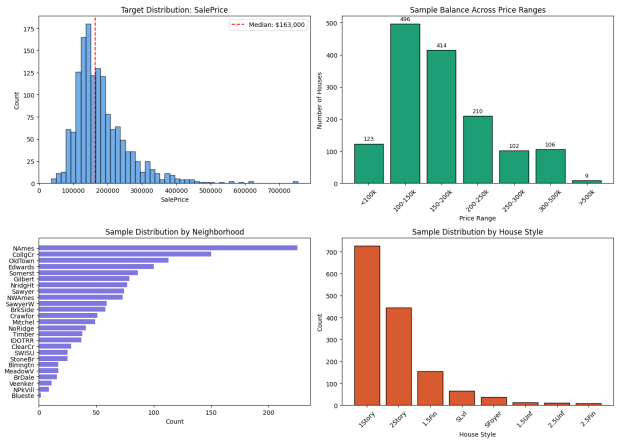

Fig. 2. Class Balance of house Sale Price

C. Feature Correlation

From Figure 3, we can see the top 10 most important features. OverallQual, measurements like square footage, and the newness of the house being built are features that are most correlated with SalesPrice. This is obvious, but this is shown through the training process in which data is trended to see with features have stronger and weaker linear trend with housing price. This can be see in feature 4.

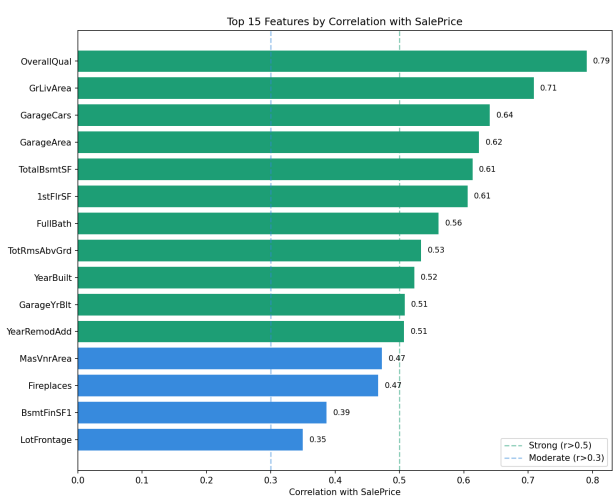

Fig. 3. Top 10 features most correlated with SalePrice. OverallQual shows the strongest correlation (r=0.79), followed by GrLivArea (r=0.71).

III. Proposed Approach

A. Logistic Regression

We chose to work with logistic regression as our first model, just to get a good baseline from its accuracy scores. We predict that logistic regression won't perform extremely well due to the nature of the dataset, but it is still a good starting point in our eyes.

B. Random Forest

For our next model, we have decided to use Random Forest. A Random Forest model will likely be more accurate than logistic regression, given it uses random subsets of data to train for variance in pricing due to specific features. We also plan to implement feature selection into this model. Our dataset has 163 features, so dimensionality reduction is necessary to prevent overfitting.

C. XGBoost

For our final model, we have decided to use XGBoost, where Random Forests builds trees independently and averages them out, and each tree is trained to correct the errors of the previous one, gradually improving the prediction accuracy with each iteration. This makes it well suited for a dataset like ours, where housing prices are influenced by complex, non-linear relationships between features. We also expect XGBoost to benefit from the feature selection applied in the previous step, as reducing noise in the dataset should allow the model to focus its boosting process on only the important patterns. We predict XGBoost will be our strongest performer given its ability to handle structured tabular data and its track record in regression tasks similar to housing price predictions (like forex, commodities trading, etc).

IV. Team Roles

A. Stefan Smith

Stefan will write the documents; he was responsible for writing sections I, III, and IV of this report. He will also work on the XGBoost model training and feature evaluation.

B. Matthew Ferrari

Matthew generated all figures and writing for section II, and was responsible for data preprocessing.

C. Faraz Ghouri

Matthew and Faraz have both worked on EDA and contributed to the figures and explanations on that end.

D. Dennis Madapatu

Dennis will work on training the Random Forest model and feature importance evaluation

E. Begad Alymany

Begad will work on data pre-processing, feature selection, and feature extraction for the XGBoost and other models.

F. Alex Worthington

Alex will work on team coordination and a fourth model type, which we have yet to decide on. It will likely be some kind of neural network.

V. Tentative Schedule

- Week 1 - Work on preprocessing, feature selection, dimensionality reduction
- Week 2 - Model training for all model types.
- Week 3 - Feature importance analysis, debugging model creation if needed.
- Week 4 - Put together presentation and final report.